

SYSTEM REQUIREMENT SPECIFICATION (SRS)

PROJECT TITLE: Blast Assist, Safety Blasting Arrangement App

GROUP NAME:

MEMBERS:

Safety Blasting App

An app that help blasting workers to calculate blasting materials fast for a specific mine. It also target community members by informing them the next blasting is going to occur for them to stay on the safe line.

1. INTRODUCTION

1.1 Purpose of this document

This document defines the functional and non-functional requirements for the BlastAssist mobile application. It serves as the technical reference for development, testing, and client acceptance.

1.2 Scope

What the app does:

- Calculate explosive type, burden, spacing, hole depth, detonator quantity and stemming (gravel/soil) based on rock properties and blast area geometry.
- Estimates fragmentation throw distance and explosive danger level.
- Notifies community members of blast times and distances.

What the app does not do:

- Denote explosives remotely
- Replace on-site blast engineer judgement.
- Track inventory of explosives.

1.3 Engineering domain and problem statement

Domain: Mining Engineering/ Blasting safety

Problem: Blasting workers currently estimate materials manually, leading to inefficiency and safety risks. Community members near mines lack real-time blast alerts. This app automates calculations and provides location based safety notifications.

2. SYSTEM OVERVIEW

2.1 System description

A React Native/ Expo app that accepts rock density, rock strength, blast area size, fracture point, water content and elasticity. It returns a blast design: explosive type, burden, spacing, hole depth detonator needs, stemming amount, danger level, and fragment throw distance. Separately, the app lets mine operators set blast times and notify registered community members within a safety radius.

2.2 Target users

User Technical level Environment Blasting worker Medium (field training) Mine site, sometimes offline Community member Low home, expects simple alerts

2.3 Technology overview

- Frontend: Expo + React Native
- Backend & DB: Firebase Fire store
- Auth: Firebase Authentication (email/password)
- Storage: Firebase Storage (blast plan images, if any)
- Notifications: Firebase Cloud Messaging

2.4 Firebase data model (preliminary)

Collection: blast calculations

- User ID(string)

- Input data (map: density, strength, area size, fracture, water, elasticity)
- Results (map: explosive type, burden, spacing, hole depth, detonators, stemming amount, danger level, frag distance)
- Time stamp

Collection: blast schedules

- Operator ID (string)
- Blast date time (timestamp)
- Location (geo point)
- Safety radius meters (number)
- Notification sent (Boolean)

Collection: community members

- User ID (string)
- Home location (geo point)
- Alert radius (number)

2.5 Assumptions and constraints

- Internet required for Firebase sync (but calculations can work offline if logic is local).
- GPS must be enabled for community alerts.
- Android 10+ (API 29) minimum.
- All calculations are estimates, final blast design requires a licensed engineer.

3. Functional Requirements

ID Requirement Statement Priority Firebase Service

- FR-001 The system shall allow blasting worker to register/login with email/password. High Firebase Auth
- FR-002 The system shall accept rock density, rock strength, blast area size, fracture point, water content, and elasticity as inputs. High local state.
- FR-003 The system shall calculate and display explosive type (e.g., ANFO, emulsion). High Fire store (store result)
- FR-004 The system shall calculate burden, spacing, hole depth, and number of detonators. High Fire store.

- FR-005 The system shall calculate amount of oil/gravel for stemming. High Fire store
- FR-006 The system shall calculate a danger level (Low/Medium/High) based on explosive type and water content. High Fire store.
- FR-007 The system shall calculate estimated fragmentation throw distance in meters. High Fire store.
- FR-008 The system shall allow a mine operator to schedule a blast with time, location, and safety radius. High Fire store.
- FR-009 The system shall send a push notification to community members within the blast radius at least 1 hour before blast time. High FCM + Fire store.
- FR-010 The system shall allow a community member to register their home location and preferred alert radius. Medium Fire store + Geo queries
- FR-011 The system shall save every blast calculation to the user's history, ordered by date. Medium Fire store
- FR-012 The system shall display a simple safety distance map for community members before a blast. Medium Fire store + Map view

4. Non-Functional Requirements

- Performance: Calculation results shall appear within 2 seconds after input.
- Security: Fire store rules shall restrict reads/writes to authenticated users only; community members may only read blast schedules, not calculations.
- Usability: A blasting worker shall complete a full calculation in 5 steps or fewer.
- Compatibility: Android 10+ (API 29) and iOS 14+.
- Reliability: If offline, the app shall save input data locally and sync when internet returns.
- Safety disclaimer: The app must display a warning: "Final blast design must be verified by a qualified engineer."